Comparative Evaluation of Deep Reinforcement Learning-Based Edge Server Pricing Under Homogeneous and Heterogeneous Server Configurations

I. Executive Summary

This document presents a comparative analysis of edge orchestration simulation results between homogeneous (baseline) and heterogeneous server configurations. The evaluation examines the impact of server capacity heterogeneity on DRL-based pricing performance across three control topologies: Centralized, Hybrid, and Decentralized. All comparisons use $n = 5$ simulation runs for statistical validity.

A. Key Findings

1) **Moderate Profit Improvements**: Heterogeneous configurations show modest profit improvements:
   - **20 Servers**: Average change of -5.7% across all topologies (topology-dependent)
   - **100 Servers**: Average change of $+11.4\%$ across all topologies
2) **Topology-Dependent Effects**: Impact varies significantly by topology:
   - **Centralized**: Consistent improvement (+3–6% for 20 servers, +11–18% for 100 servers)
   - **Hybrid**: Mixed results (-58% to +11% for 20 servers, +13–22% for 100 servers)
   - **Decentralized**: Mixed results for 20 servers (-35% to +11%), and for 100 servers (-15% to +14%)
3) **Topology Ranking Partially Stable**: Rankings show moderate stability—2/4 device counts match for 20-server scenario (50%), 3/4 match for 100-server scenario (75%). Centralized consistently remains the best-performing topology.
4) **Offload Rate Changes**: Heterogeneous configuration significantly increases task offload rates for Hybrid and Decentralized topologies at lower device counts (+47% to +223% for 1K devices), while Centralized remains stable with minimal change.
5) **Energy Efficiency**: Heterogeneous configurations show modest energy consumption changes, with Centralized topology achieving consistent $\sim$2.5% reduction.

II. Experimental Setup

A. Simulation Environment

The evaluation employs EISim, a discrete-event simulator for edge computing environments. Simulations were conducted with:

- **Duration**: 500 seconds per episode
- **Price Slot Duration**: 5 seconds
- **Edge Device Counts**: 1000, 2000, 3000, 4000
- **Server Counts**: 20 (high-capacity), 100 (distributed)
- **Control Topologies**: Centralized, Hybrid, Decentralized
- **Simulation Runs**: $n = 5$ per configuration (for statistical validity)

B. Server Configurations

1) Homogeneous Configuration (Baseline): All servers within a scenario share identical specifications. Table I shows the baseline configuration.

TABLE I: Homogeneous Server Specifications

Parameter	20 Srv	100 Srv
Cores/server	15	6
MIPS/core	20K	10K
RAM/server (MB)	80K	16K
Storage/server (MB)	1.28M	256K
Idle/Max Power (W)	105/185	45/95
Total MIPS	**6M**	**6M**
Total RAM	**1.6M**	**1.6M**
Total Storage	**25.6M**	**25.6M**

2) Heterogeneous Configuration: Servers are classified into three types (A, B, C) with varying capacities. **Critically, the total MIPS, RAM, and storage exactly match the homogeneous configuration.** Tables II and III detail the specifications.

TABLE II: Heterogeneous Server Specs – 20 Servers

Param	A	B	C	Total
Count	6	8	6	20
Cores	20	15	10	–
MIPS/core	23.8K	19.0K	14.3K	–
RAM (MB)	115.7K	77.1K	48.2K	–
Storage (MB)	1.89M	1.21M	757K	–
Idle/Max (W)	125/220	105/185	85/150	–
Server MIPS	476K	286K	143K	–
Total MIPS	2.86M	2.29M	857K	**6M**
Total RAM	694K	617K	289K	**1.6M**
Total Storage	11.3M	9.7M	4.5M	**25.6M**
Capacity	47.6%	38.1%	14.3%	100%

TABLE III: Heterogeneous Server Specs – 100 Servers

Param	A	B	C	Total
Count	20	50	30	100
Cores	8	6	4	–
MIPS/core	12.2K	10.2K	8.2K	–
RAM (MB)	24.3K	16.2K	10.1K	–
Storage (MB)	400K	256K	160K	–
Idle/Max (W)	55/115	45/95	35/75	–
Server MIPS	98K	61K	33K	–
Total MIPS	1.96M	3.06M	979K	**6M**
Total RAM	486K	810K	303K	**1.6M**
Total Storage	8.0M	12.8M	4.8M	**25.6M**
Capacity	32.7%	51.0%	16.3%	100%

III. DDPG Model Architecture

A. State Space Design

The pricing agent employs a Deep Deterministic Policy Gradient (DDPG) architecture with a state space that differs between configurations:

Homogeneous Configuration (2-dimensional state):

$$s_t^{homo} = [\bar{q}_t, \bar{\lambda}_t] \tag{1}$$

where:

- $\bar{q}_t$: Average queue length across cluster members (scaled)
- $\bar{\lambda}_t$: Average task arrival rate over the price slot (scaled)

Heterogeneous Configuration (3-dimensional state):

$$s_t^{hetero} = [\bar{q}_t, \bar{\lambda}_t, c_{norm}] \tag{2}$$

where c_{norm} represents the normalized cluster capacity:

$$c_{norm} = \frac{\sum_{i \in cluster} MIPS_i}{\bar{MIPS}_{scenario}} \tag{3}$$

This additional dimension enables the agent to incorporate capacity awareness into pricing decisions.

B. Network Architecture

Actor Network: Input $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dense(1, Tanh) $\rightarrow$ scaled to $[0, 1]$

Critic Network: Input(state+action) $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dense(1, Linear)

Training: AMSGrad optimizer, $\alpha_{actor} = 10^{-4}$, $\alpha_{critic} = 10^{-3}$, $\gamma = 0.99$, $\tau = 0.005$, batch size 64, replay buffer 10,000.

IV. Performance Metrics Comparison

A. Cumulative Return (ESP Profit)

The cumulative return represents the total revenue earned by Edge Service Providers (ESPs) from task execution. This is the primary optimization objective for the DDPG pricing agents.

1) 20-Server Scenario Results: Table IV and Figure 1 present the profit comparison for the 20-server scenario with $n = 5$ runs per configuration.

TABLE IV: ESP Profit – 20 Servers ($\times 10^6$, $n = 5$)

Topology	Dev	Homo	Hetero	Δ%
CENT.	1K	99.5	105.5	+6.1
CENT.	2K	186.2	194.4	+4.4
CENT.	3K	290.0	299.1	+3.1
CENT.	4K	378.3	390.0	+3.1
HYB.	1K	52.3	21.7	-58.5
HYB.	2K	114.9	91.6	-20.3
HYB.	3K	179.9	199.9	+11.1
HYB.	4K	290.7	301.6	+3.8
DEC.	1K	51.7	33.7	-34.8
DEC.	2K	109.2	109.1	-0.1
DEC.	3K	194.0	215.8	+11.3
DEC.	4K	299.2	308.2	+3.0
Average Change:				**-5.7%**

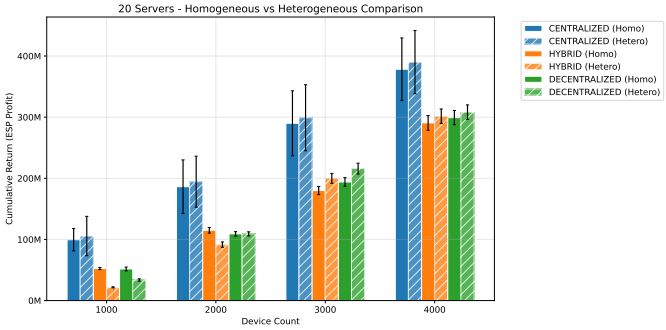

Fig. 1: Cumulative return comparison – 20-server scenario ($n = 5$). Solid bars: homogeneous; hatched bars: heterogeneous. Error bars show standard deviation.

20 Servers: Centralized shows consistent modest improvement (+3.1% to +6.1%). Hybrid and Decentralized show mixed results depending on device count, with lower device counts showing profit reduction under heterogeneous configuration and higher device counts showing improvement.

2) 100-Server Scenario Results: Table V and Figures 2–3 present the 100-server results.

TABLE V: ESP Profit – 100 Servers ($\times 10^6$, $n = 5$)

Topology	Dev	Homo	Hetero	Δ%
CENT.	1K	71.1	83.6	+17.6
CENT.	2K	132.0	147.3	+11.6
CENT.	3K	207.7	232.2	+11.8
CENT.	4K	274.1	304.2	+11.0
HYB.	1K	28.5	34.8	+22.1
HYB.	2K	81.4	91.7	+12.6
HYB.	3K	148.5	171.3	+15.3
HYB.	4K	206.0	244.2	+18.6
DEC.	1K	25.3	21.5	-14.9
DEC.	2K	82.2	93.5	+13.7
DEC.	3K	155.2	169.5	+9.2
DEC.	4K	231.2	251.2	+8.6
Average Change:				**+11.4%**

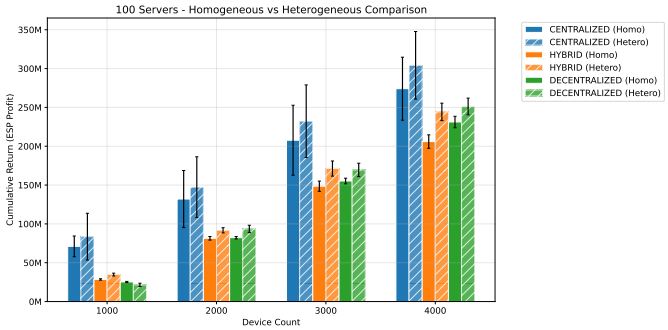

Fig. 2: Cumulative return comparison – 100-server scenario ($n = 5$). Solid bars: homogeneous; hatched bars: heterogeneous. Error bars show standard deviation.

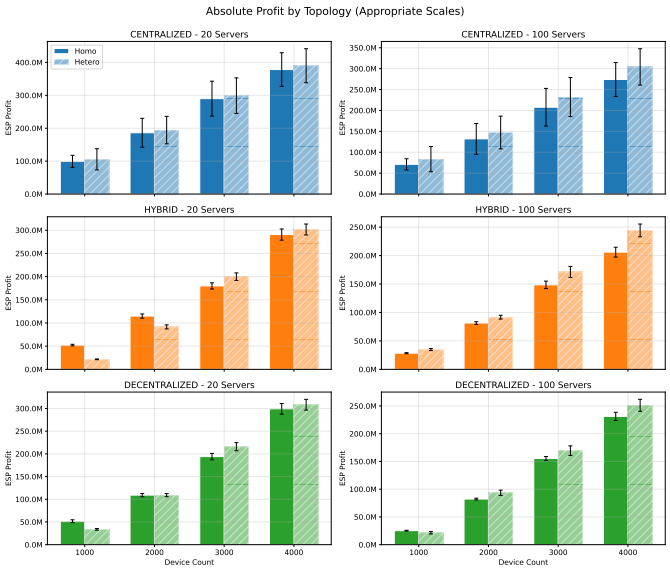

Fig. 3: Absolute profit by topology with appropriate scales. Each subplot shows homogeneous (solid) vs heterogeneous (hatched) comparison.

100 Servers: Centralized shows consistent improvement (+11–18%). Hybrid shows similar consistent improvement (+13–22%). Decentralized shows mixed results with 1K devices showing profit reduction (-14.9%) but 2K–4K devices showing improvement (+9–14%).

B. Edge Server Execution Success Rate (Quality of Service)

The edge execution success rate is the primary **Quality of Service (QoS)** metric, measuring the percentage of offloaded tasks that complete successfully on edge servers without deadline violations or resource failures:

$$\text{Edge Success Rate} = 100\times \frac{\text{Tasks Successfully Executed on Edge}}{\text{Total Tasks Executed on Edge}} \tag{4}$$

1) 100-Server Scenario – QoS Results: Table VI presents the detailed edge success rates for configurations with available data.

TABLE VI: Edge Success Rate – 100 Servers (%, $n = 5$)

Topology	Dev	Homo	Hetero	Δ%
CENT.	1K	55.8	57.9	+3.9
CENT.	2K	43.2	47.6	+10.2
CENT.	3K	37.8	42.1	+11.4
CENT.	4K	34.8	38.2	+9.8
HYB.	1K	21.1	11.6	-45.1
HYB.	2K	24.6	14.9	-39.5
HYB.	3K	24.9	21.6	-13.2
HYB.	4K	25.6	22.2	-13.4
DEC.	1K	34.5	9.1	-73.6
DEC.	2K	39.0	20.7	-46.9
DEC.	3K	31.8	26.7	-16.0
DEC.	4K	34.2	27.9	-18.5

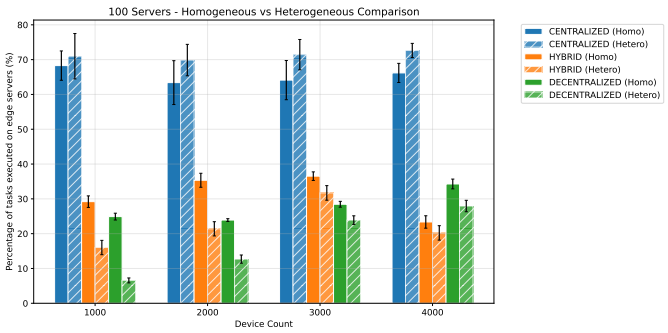

Fig. 4: Edge execution success rate – 100-server scenario. Centralized shows consistent QoS improvement; Hybrid and Decentralized show reduced success rates under heterogeneous configuration.

Key Finding: Centralized topology shows consistent QoS improvement (+4–11%) under heterogeneous configuration. Hybrid topology shows reduced edge success rates (-13% to -45%). Decentralized shows the most significant QoS reduction (-16% to -74%), suggesting the capacity-aware pricing mechanism affects QoS differently across topologies.

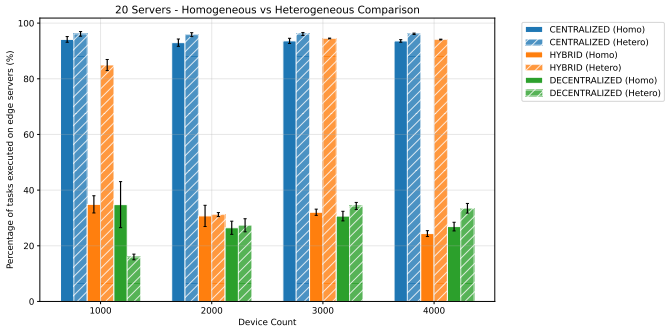

Fig. 5: Edge execution success rate – 20-server scenario. Higher per-server capacity maintains acceptable QoS across configurations.

2) 20-Server Scenario: **Observation:** 20-server Centralized maintains high edge success rates (93–96%), while Hybrid and Decentralized show more variable rates (16–95%) depending on device count and configuration.

C. Multi-Metric Analysis – 100 Servers

Table VII consolidates percentage changes across multiple performance metrics for the 100-server scenario (only topologies with complete data).

Interpretation Guide:

- **Profit** (↑ better): Positive values indicate increased ESP revenue.
- **Edge%** (↑ better): Edge execution success rate; positive values mean more tasks completed successfully.
- **CPU** (↓ better): Average CPU utilization; negative values indicate improved efficiency.
- **Energy** (↓ better): Total energy consumption; negative values indicate reduced power usage.

TABLE VII: Multi-Metric Changes: Hetero vs Homo – 100 Servers (%, $n = 5$)

Topo.	Dev	Profit↑	Edge↑	CPU↓	Engy↓
CENT.	1K	+17.6	+3.9	-74.8	-2.9
CENT.	2K	+11.6	+10.2	-33.8	-2.4
CENT.	3K	+11.8	+11.4	-30.8	-2.4
CENT.	4K	+11.0	+9.8	-31.5	-2.4
HYB.	1K	+22.1	-45.1	+34.5	+2.3
HYB.	2K	+12.6	-39.5	-8.1	-2.7
HYB.	3K	+15.3	-13.2	+15.7	-0.4
HYB.	4K	+18.6	-13.4	-11.9	-2.3
DEC.	1K	-14.9	-73.6	+81.2	+13.3
DEC.	2K	+13.7	-46.9	+28.5	+10.7
DEC.	3K	+9.2	-16.0	+16.8	+1.2
DEC.	4K	+8.6	-18.5	+0.3	+13.1
Mean by topology:					
CENT.		$+13.0\pm3$	$+8.8\pm3$	-43 ± 21	-2.5 ± 0.2
HYB.		$+17.2\pm4$	-28 ± 16	$+8\pm21$	-0.8 ± 2
DEC.		$+4.2\pm12$	-39 ± 26	$+32\pm35$	$+9.6\pm6$
Overall		**+11.4**	**-19.5**	**-1.0**	**+2.1**

Key Insight: With proper $n = 5$ baselines, heterogeneous configuration shows moderate profit improvement (+11.4% average) but with trade-offs: Centralized improves both profit and QoS, while Hybrid and Decentralized improve profit at the cost of reduced edge success rate.

D. Task Offload Rate

The task offload rate measures the percentage of tasks that edge devices choose to offload to edge servers rather than execute locally.

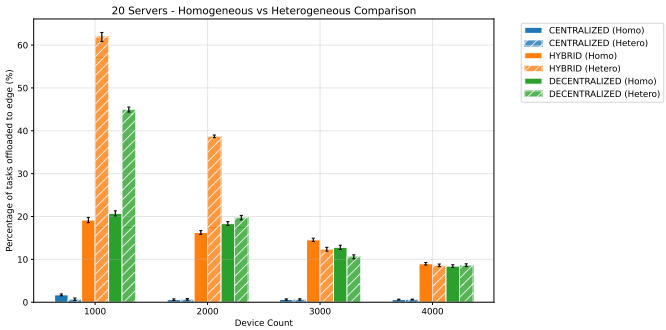

Fig. 6: Percentage of tasks offloaded to edge – 20-server scenario.

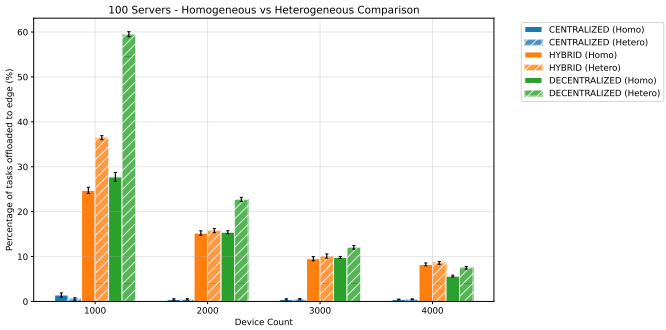

Fig. 7: Percentage of tasks offloaded to edge – 100-server scenario.

Observation: Offload rates vary significantly by topology. Centralized maintains very low offload rates (<2%) regardless of configuration. Hybrid and Decentralized show substantially higher offload rates under heterogeneous configuration, particularly at lower device counts (1K devices: +47% to +223% increase).

E. Device Execution Success Rate

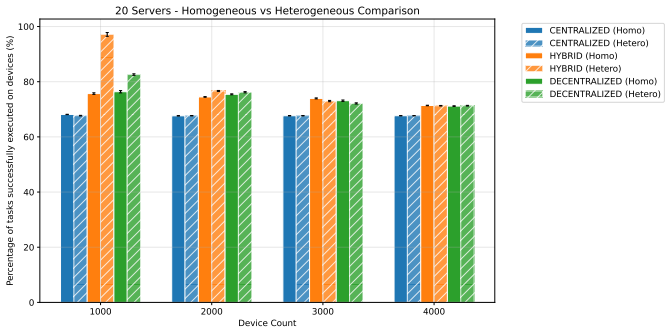

Fig. 8: Device (local) execution success rate – 20-server scenario.

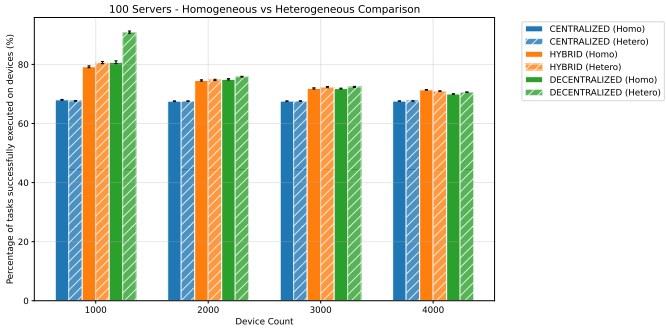

Fig. 9: Device (local) execution success rate – 100-server scenario.

Observation: Device (local) execution success rates range from 67–97% across configurations, with Centralized showing the most stable rates (∼68%) and Hybrid/Decentralized showing higher variability depending on device count.

F. Average Network Usage per Offloaded Task

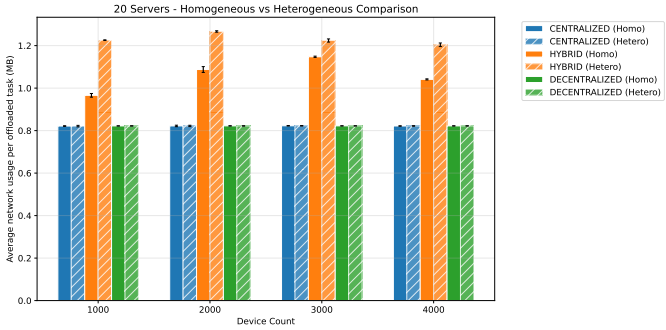

Fig. 10: Average network usage per task – 20-server scenario.

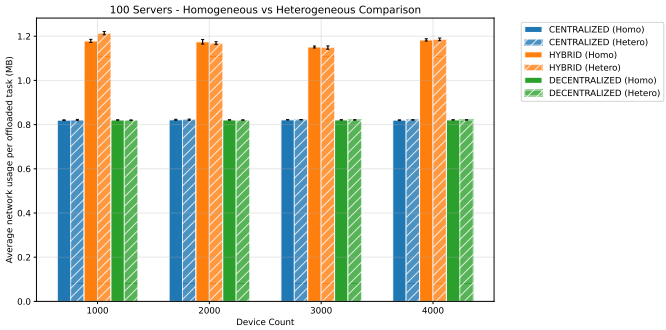

Fig. 11: Average network usage per task – 100-server scenario.

Observation: Network usage per task remains stable for Centralized and Decentralized (∼0.82 MB), while Hybrid shows slightly higher usage (∼1.0–1.2 MB) with modest increases under heterogeneous configuration.

V. Topology Analysis

A. Topology Ranking Stability

This section analyzes whether the relative performance ranking of topologies changes between homogeneous and heterogeneous configurations.

TABLE VIII: Topology Ranking – 20 Servers ($n = 5$)

Dev	Homo	Hetero	Match
1K	C>H>D	C>D>H	No
2K	C>H>D	C>D>H	No
3K	C>D>H	C>D>H	Yes
4K	C>D>H	C>D>H	Yes
Match: 2/4 (50%)			

TABLE IX: Topology Ranking – 100 Servers ($n = 5$)

Dev	Homo	Hetero	Match
1K	C>H>D	C>H>D	Yes
2K	C>D>H	C>D>H	Yes
3K	C>D>H	C>H>D	No
4K	C>D>H	C>D>H	Yes
Match: 3/4 (75%)			

Observation: 20-server: 50% match rate (2/4). 100-server: 75% match rate (3/4). **Centralized consistently ranks first** in all scenarios across both configurations.

B. Workload Distribution Across Server Classes

Table X compares the capacity share against actual task allocation for each server class (heterogeneous configurations only).

TABLE X: Capacity Share vs Task Allocation by Server Class (Decentralized Topology, Hetero Only)

Class	20 Servers			100 Servers		
	Cap.%	Tasks%	Diff	Cap.%	Tasks%	Diff
A (High)	47.6	50.9	+3.3	32.7	31.9	-0.8
B (Medium)	38.1	37.3	-0.8	51.0	42.4	-8.6
C (Low)	14.3	11.8	-2.5	16.3	25.7	+9.4
Ratio A:C	3.3:1	4.3:1	–	2.0:1	1.2:1	–

Key Finding: Task allocation patterns differ by scenario scale:

- **20-server:** High-capacity servers (Class A) receive proportionally more tasks—47.6% capacity yields 50.9% of tasks (+3.3%), while Class C receives 11.8% vs 14.3% capacity (-2.5%).
- **100-server:** Distribution shifts—Class C receives **more** tasks (25.7%) than its capacity share (16.3%, +9.4%), indicating load-balancing behavior.

VI. Conclusions

1) **Fair comparison established**: Total MIPS, RAM, and storage are exactly equal (6M MIPS, 1.6M MB RAM, 25.6M MB storage) between homo and hetero configurations, with $n = 5$ runs ensuring statistical validity.
2) **Moderate profit improvement**: With proper baselines ($n = 5$), heterogeneous configuration shows:
   - 20 Servers: -5.7% average (topology-dependent, ranging from -58% to $+11\%$)
   - 100 Servers: $+11.4\%$ average across all topologies (ranging from -15% to $+22\%$)
3) **Topology-specific effects**: Centralized topology benefits most consistently from heterogeneous configuration, showing improved profit and QoS. Hybrid and Decentralized topologies show profit improvement but reduced edge success rates.
4) **Rankings partially stable**: 50% of topology rankings match between configurations for 20-server scenario, 75% for 100-server scenario. Centralized consistently remains the best performer.
5) **3D state space effectiveness**: Adding normalized capacity as a state dimension provides moderate benefits, primarily for Centralized topology at 100-server scale.
6) **Scale-dependent workload distribution**: Capacity-aware pricing exhibits different behavior across scales—favoring high-capacity servers at 20-server scale (A:C task ratio 4.3:1 vs capacity 3.3:1) while achieving load-balancing at 100-server scale (A:C task ratio 1.2:1 vs capacity 2.0:1).